

# Laboratory 2 Report Template:

## Signals and Systems

### Signal Transformations in Hardware

#### Time Shift, Amplitude Scaling, and Time Reversal

### Graded Submission Requirement

**Deliverable (1/11):** This is a graded assignment. Include your name and student ID in every required screenshot and in as many additional screenshots as possible. Screenshots without clear student identification may not receive full credit.

**Checkpoint (1/7):** Before submission, verify that each screenshot is legible and contains visible labels/measurements where required.

### Student Information

|         |       |             |       |
|---------|-------|-------------|-------|
| Name 1  | _____ | Name 2      | _____ |
| NetID 1 | _____ | NetID 2     | _____ |
| Section | _____ | Date        | _____ |
| TA      | _____ | Random Seed | _____ |

### Note on Prompts

- **Pre-lab Deliverable ( $N$ ):** Pre-lab work to complete before lab.
- **Checkpoint ( $N$ ):** Items to show your TA in lab.
- **Discussion ( $N$ ):** Analysis and interpretation in your report.
- **Deliverable ( $N$ ):** Required report artifacts and evidence.

## 1 Assigned Parameters

Table 1. TA-Assigned Base Signal Parameters

| Parameter   | Description                                               | Assigned Value |
|-------------|-----------------------------------------------------------|----------------|
| $A$         | Base signal amplitude (V)                                 | _____          |
| $f_0$       | Fundamental frequency (Hz)                                | _____          |
| $\phi$      | Initial phase (rad)                                       | _____          |
| $\alpha$    | Exponential decay rate ( $\text{s}^{-1}$ ), if applicable | _____          |
| Signal Type | Sinusoidal / Exponential / Damped Sinusoidal              | _____          |
| $t_0$       | Time shift duration (ms)                                  | _____          |
| $B$         | Amplitude scale factor                                    | _____          |

## 2 Pre-Lab Results

**Pre-lab Deliverable (1/1):** Attach your complete pre-lab pages or include concise summaries below.

### 2.1 Transformation Derivations

- $y_1(t) = x(t - t_0)$ : \_\_\_\_\_
- $y_2(t) = B x(t)$ : \_\_\_\_\_
- $y_3(t) = x(-t)$ : \_\_\_\_\_
- $y_4(t) = B x(-t - t_0)$ : \_\_\_\_\_

### 2.2 Discrete-Time Mapping

- Shift depth:  $k = \lfloor t_0/T_s \rfloor =$  \_\_\_\_\_ samples
- MATLAB mapping for  $y_1, y_2, y_3, y_4$ : \_\_\_\_\_

### 2.3 Sketch Summary

Insert pre-lab sketch image (x, y1, y2, y3, y4) or attach scanned page

### 3 Phase 1: Base Signal Acquisition

**Checkpoint (2/7):** Verify base signal generation and acquisition in both scope and MATLAB.

#### 3.1 Step 1.1 Base Signal on Oscilloscope

**Deliverable (2/11):** Submit scope screenshot with name/ID, time/div, volts/div, measured frequency, measured amplitude.

Table 2. Measured Base Signal Parameters

|                                     |       |
|-------------------------------------|-------|
| Measured frequency (Hz)             | _____ |
| Measured peak-to-peak voltage (Vpp) | _____ |
| Time/div                            | _____ |
| Volts/div                           | _____ |

**Step 1.1 Evidence: Oscilloscope screenshot (base signal)**

#### 3.2 Step 1.2 MATLAB Base Signal Acquisition

**Deliverable (3/11):** Submit MATLAB plot of x\_hw with labeled axes and title.

**Step 1.2 Evidence: MATLAB base-signal plot screenshot**

**Discussion (1/7):** Compare scope and MATLAB base signal. Report amplitude error and likely cause.

$$\varepsilon_A(\%) = \frac{|A_{scope} - A_{MATLAB}|}{A_{scope}} \times 100 = \underline{\hspace{2cm}}\%$$

## 4 Phase 2: Hardware Transformations

### 4.1 Step 2.1 Amplitude Scaling

**Checkpoint (3/7):** Implement gain control with potentiometer and verify B ratio. **Deliverable (4/11):** Submit dual-channel scope screenshot (scaled vs original) with annotated ratio.

Table 3. Amplitude Scaling Measurements

|                              |       |
|------------------------------|-------|
| $A_{Ch1}$ (scaled)           | _____ |
| $A_{Ch2}$ (original)         | _____ |
| $B_{meas} = A_{Ch1}/A_{Ch2}$ | _____ |
| Assigned $B$                 | _____ |
| Percent error in $B$         | _____ |

**Step 2.1 Evidence: Scope screenshot for amplitude scaling**

**Discussion (2/7):** Why is potentiometer reading performed outside ISR? Why must shared gain variable be volatile?

### 4.2 Step 2.2 Time Shift

**Checkpoint (4/7):** Implement circular buffer delay with depth k and verify time shift. **Deliverable (5/11):** Submit dual-channel scope screenshot with cursor-measured delay.

Table 4. Time Shift Measurements

|                      |       |
|----------------------|-------|
| Predicted $t_0$ (ms) | _____ |
| Measured delay (ms)  | _____ |
| Absolute error (ms)  | _____ |
| Percent error        | _____ |

**Step 2.2 Evidence: Scope screenshot for time shift**

**Discussion (3/7):** Explain time delay vs phase shift equivalence for sinusoid and why it does not generalize to all signal classes.

#### 4.3 Step 2.3 Time Reversal

**Checkpoint (5/7):** Implement reversed LUT indexing and verify waveform reversal. **Deliverable (6/11):** Submit dual-channel scope screenshot with key distinguishing feature annotated.

**Step 2.3 Evidence: Scope screenshot for time reversal**

**Discussion (4/7):** Classify  $x(t)$  as even/odd/neither and explain expected reversal behavior.

#### 4.4 Step 2.4 Combined Transformation

**Checkpoint (6/7):** Implement and verify  $y_4(t) = B x(-t - t_0)$ . **Deliverable (7/11):** Submit dual-channel scope screenshot with ratio and time-offset annotations.

**Step 2.4 Evidence: Scope screenshot for combined transformation**

## 5 Phase 3: MATLAB Verification

### 5.1 Step 3.1 Acquire All Signals

**Checkpoint (7/7):** Acquire  $x$ ,  $y_1$ ,  $y_2$ ,  $y_3$ ,  $y_4$  into MATLAB. **Deliverable (8/11):** Submit five MATLAB plots with proper labeling.

**Step 3.1 Evidence: MATLAB screenshots for  $x$ ,  $y_1$ ,  $y_2$ ,  $y_3$ ,  $y_4$**

### 5.2 Step 3.2 Predicted vs Measured Overlays

**Deliverable (9/11):** Submit four overlays: base, hardware, predicted for each transformation.

**Step 3.2 Evidence: MATLAB overlay screenshots ( $y_1$ - $y_4$ )**

### 5.3 Step 3.3 NRMSE Quantification

**Deliverable (10/11):** Report NRMSE for  $y_1$ ,  $y_2$ ,  $y_3$ ,  $y_4$  using the specified formula.

Table 5. NRMSE Summary

|                                       |       |
|---------------------------------------|-------|
| Time Shift ( $y_1$ ) NRMSE (%)        | _____ |
| Amplitude Scaling ( $y_2$ ) NRMSE (%) | _____ |
| Time Reversal ( $y_3$ ) NRMSE (%)     | _____ |
| Combined ( $y_4$ ) NRMSE (%)          | _____ |



**Step 3.3 Evidence: MATLAB output/table screenshot for NRMSE**

**Discussion (5/7):** Which transformation has largest NRMSE and why? Separate systematic vs random error sources.

## 6 Data Analysis

### 6.1 Transformation Summary

Table 6. Transformation Summary Table

| Parameter                          | Theoretical | Measured | % Error |
|------------------------------------|-------------|----------|---------|
| Base amplitude $A$ (V)             |             |          |         |
| Base frequency $f_0$ (Hz)          |             |          |         |
| Scaled amplitude $B \cdot A$ (V)   |             |          |         |
| Gain ratio $B_{meas}$              |             |          |         |
| Time shift $t_0$ (ms)              |             |          |         |
| Phase shift $\phi_0$ (rad)         |             |          |         |
| Combined amplitude $B \cdot A$ (V) |             |          |         |
| Combined shift $t_0$ (ms)          |             |          |         |

**Discussion (6/7):** Identify the largest discrepancy source for each transformation and justify with measured evidence.

### 6.2 Symmetry and Boundary Effects

**Discussion (7/7):** Classify signal symmetry (even/odd/neither) and discuss `circshift` wrap-around artifacts with mitigation strategy.

### 6.3 Error Source Ranking

**Deliverable (11/11):** Rank top three contributors to error using NRMSE and physical justification (PWM quantization, RC phase distortion, ADC quantization, UART jitter, potentiometer noise, ISR jitter, etc.).

## Final Submission Checklist

- ☐ Pre-lab derivations and sketches included.
- ☐ Step 1.1 scope screenshot with name/ID.
- ☐ Step 1.2 MATLAB base signal plot.
- ☐ Step 2.1-2.4 scope screenshots with annotations.
- ☐ Step 3.1 five MATLAB signal plots.
- ☐ Step 3.2 four overlay plots.
- ☐ Step 3.3 NRMSE table/screenshot.
- ☐ Completed analysis tables and discussion responses.